$\Rightarrow (15^{23} + 23^{23})$ is divisible by 38 and hence by 19

$\Rightarrow$ On dividing $(15^{23} + 23^{23})$ by 19, we get 0 as remainder.

338. When n is even, $(x^n - a^n)$ is divisible by $(x + a)$.

Now, $2^{256} = (2^4)^{64} = (16)^{64}$.

$\therefore (16^{64} - 1^{64})$ is divisible by $(16 + 1)$

$\Rightarrow (16^{64} - 1)$ is divisible by 17

$\Rightarrow (2^{256} - 1)$ is divisible by 17

$\Rightarrow$ On dividing 2^{256} by 17, we get 1 as remainder.

339. When n is even, $(x^n - a^n)$ is divisible by both $(x - a)$ as well as $(x + a)$.

Now, $(7^{6n} - 6^{6n}) = [(7^3)^{2n} - (6^3)^{2n}] = [(343)^{2n} - (216)^{2n}]$.

$\therefore (7^{6n} - 6^{6n})$ is divisible by both $(7 - 6)$ and $(7 + 6)$

$\Rightarrow (7^{6n} - 6^{6n})$ is divisible by 13.

And, $[(343)^{2n} - (216)^{2n}]$ is divisible by both $(343 - 216)$ and $(343 + 216)$

$\Rightarrow (7^{6n} - 6^{6n})$ is divisible by both 127 and 559.

340. Let $2^{32} = x$. Then, $(2^{32} + 1) = (x + 1)$.

Let $(x + 1)$ be completely divisible by the natural number N.

Then, $(2^{96} + 1) = [(2^{32})^3 + 1] = (x^3 + 1) = (x + 1)(x^2 - x + 1)$, which is completely divisible by N since $(x + 1)$ is divisible by N.

341. $(2^{48} - 1) = [(2^6)^8 - 1] = (64)^8 - 1$.

When n is even, $(x^n - a^n)$ is completely divisible by both $(x - a)$ and $(x + a)$.

$\therefore (64^8 - 1^8)$ is divisible by both $(64 - 1)$ and $(64 + 1)$

$\Rightarrow (2^{48} - 1)$ is divisible by both 63 and 65.

342. $n^{65} - n = n(n^{64} - 1) = n(n^{32} - 1)(n^{32} + 1)$

$= n(n^{16} - 1)(n^{16} + 1)(n^{32} + 1)$

$= n(n^8 - 1)(n^8 + 1)(n^{16} + 1)(n^{32} + 1)$

$= n(n^4 - 1)(n^4 + 1)(n^8 + 1)(n^{16} + 1)(n^{32} + 1)$

$= n(n^2 - 1)(n^2 + 1)(n^4 + 1)(n^8 + 1)(n^{16} + 1)(n^{32} + 1)$

$= (n - 1)\,n\,(n + 1)(n^2 + 1)(n^4 + 1)(n^8 + 1)(n^{16} + 1)(n^{32} + 1)$.

Clearly, $(n - 1)$, n and $(n + 1)$ are three consecutive numbers and they have to be multiples of 2, 3 and 4 as n is odd.

Thus, the given number is definitely a multiple of 24.

343. Let $a = 55$, $b = 17$.

Then, $N = a^3 + b^3 - (a + b)^3 = a^3 + b^3 - [a^3 + b^3 + 3ab(a + b)]$

$= -3ab(a + b) = -3 \times 55 \times 17 \times 72$.

Clearly, N is divisible by both 3 and 17.

344. Taking out $\lfloor 10$ common from $[\lfloor 10 + \lfloor 11 + \ldots\ldots + \lfloor 100]$, we get this expression in the form of a multiple of $\lfloor 10$ which has zeros as its last two digits. So, the last two digits of the expression $[\lfloor 10 + \lfloor 11 + \ldots\ldots + \lfloor 100]$ are zeros.

Thus, the last two digits of N must be the last two digits of the sum $(\lfloor 1 + \lfloor 2 + \ldots\ldots + \lfloor 9)$.

Now, $\lfloor 1 + \lfloor 2 + \ldots\ldots + \lfloor 9 = 1 + 2 + 6 + 24 + 120 + 720 + 5040 + 40320 + 362880$

It has clearly 13 as the last two digits.

So, the last two digits of N are 13.

345. $168 = 2^3 \times 3 \times 7$ and $\lfloor 7 = 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1 = 7 \times 5 \times 3^2 \times 2^4 = 168 \times 30$

Hence, $\lfloor 7$ and all the factorials greater than $\lfloor 7$ are divisible by 168.

Now, $N = \lfloor 1 + \lfloor 2 + \lfloor 3 + \ldots\ldots + \lfloor 99 + \lfloor 100$

$= \lfloor 1 + \lfloor 2 + \lfloor 3 + \lfloor 4 + \lfloor 5 + \lfloor 6 +$ a multiple of 168.

So, the remainder obtained on dividing N by 168 is the same as that obtained on dividing $(\lfloor 1 + \lfloor 2 + \lfloor 3 + \lfloor 4 + \lfloor 5 + \lfloor 6)$ by 168.

Now, $\lfloor 1 + \lfloor 2 + \lfloor 3 + \lfloor 4 + \lfloor 5 + \lfloor 6 = 1 + 2 + 6 + 24 + 120 + 720$
$= 873 = (168 \times 5) + 33$.

Hence, the required remainder is 33.

346. $4^{61} = 4 \times 4^{60} = 4 \times (4^4)^{15} = 4 \times (256)^{15}$.

Now, $(x^n - a^n)$ is divisible by $(x - a)$ for all values of n.

$\therefore (256^{15} - 1)$ is divisible by $(256 - 1)$ i.e. 255 and hence by 51.

$\Rightarrow$ On dividing $(256)^{15}$ by 51, we get 1 as remainder

$\Rightarrow$ On dividing 4^{60} by 51, we get 1 as remainder

$\Rightarrow$ On dividing 4^{61} by 51, remainder obtained $= (4 \times 1) = 4$.

347. $17^{36} = (17^2)^{18} = (289)^{18}$.

Now, $[(289)^{18} - 1]$ is divisible by $(289 - 1)$, i.e. 288

$\Rightarrow (17^{36} - 1)$ is divisible by 288 and hence by 36

$\Rightarrow$ On dividing 17^{36} by 36, we get 1 as remainder.

348. When n is odd, $(x^n + a^n)$ is always divisible by $(x + a)$.

$\therefore$ Each one of $(47^{43} + 43^{43})$ and $(47^{47} + 43^{47})$ is divisible by $(47 + 43)$.

349. Product of all odd natural numbers less than 5000

$= 1 \times 3 \times 5 \times 7 \times \ldots\ldots \times 4999$

$$= \frac{1 \times 2 \times 3 \times 4 \times 5 \times 6 \times \ldots\ldots \times 5000}{2 \times 4 \times 6 \times 8 \times \ldots\ldots \times 5000}$$

$$= \frac{5000!}{2^{2500}(1 \times 2 \times 3 \times 4 \times \ldots\ldots \times 2500)} = \frac{5000!}{2^{2500}.\,2500!}.$$

350. The pages of the book may be divided into 10 groups: $(1 - 100)$, $(101 - 200)$, $(201 - 300)$,..........., $(901 - 1000)$.

Clearly, for the first group, one needs 11 zeros.

For second to ninth groups, one needs 20 zeros each.

For the tenth group, one needs 21 zeros.

So, total number of zeros required $= 11 + 8 \times 20 + 21 = 192$.

351. $a^2 + b^2 + c^2 = 1$.

So, the maximum value of $a^2 b^2 c^2 = \left(\frac{1}{3} \times \frac{1}{3} \times \frac{1}{3}\right) = \frac{1}{27}$.

($\because$ when sum of three positive quantities is fixed, the product will be maximum when the quantities are equal)

Hence, maximum value of $abc = \frac{1}{\sqrt{27}} = \frac{1}{3\sqrt{3}}$.

352. Let $N = 1^5 + 2^5 + 3^5 + \ldots\ldots + (100)^5$.

Then, $N = (1^5 + 2^5 + 3^5 + \ldots\ldots + 10^5) + (11^5 + 12^5 + \ldots\ldots + 20^5) + (21^5 + 22^5 + \ldots\ldots + 30^5) + \ldots\ldots + (91^5 + 92^5 + \ldots\ldots + 100^5) = N_1 + N_2 + N_3 + \ldots\ldots + N_{10}$.

Since each one of $N_1, N_2, N_3, \ldots\ldots, N_{10}$ has the same